Behavior-Level Interpretation for Encrypted Traffic Classification via SPLT-SHAP Matching*

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Abstract

Recent encrypted traffic classification studies have adopted AI models using behavioral features such as SPLT. Although SHAP identifies important features contributing to model decisions, it remains difficult to trace these explanations back to the original packet sequence. To address this limitation, we propose SPLT-SHAP matching, which maps SHAP-derived explanations to packet-level behavioral regions in encrypted traffic. The proposed method enables intuitive behavior-level interpretation by visually linking feature-level evidence with the original SPLT sequence.

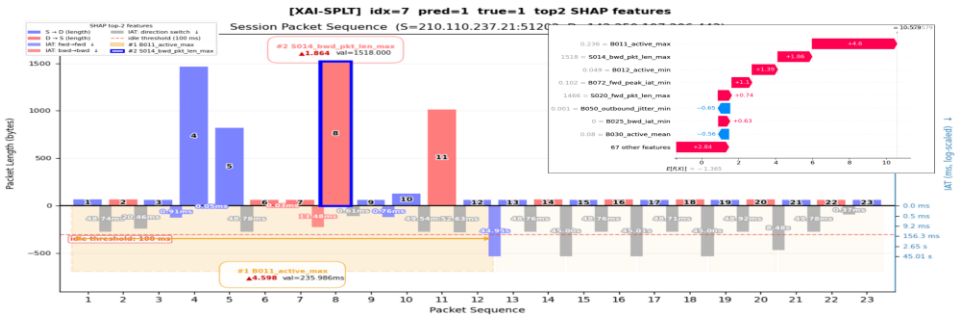
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1 Introduction

Although AI-based encrypted traffic classification achieves high performance using behavioral features such as SPLT [1], analysts still face difficulty in understanding the behavioral basis of model predictions. While SHAP [2] provides feature-level explanations, it remains difficult to trace these explanations back to the original packet sequence or traffic behavior. To address this limitation, we propose SPLT-SHAP matching, which maps SHAP-derived explanations to packet-level behavioral regions in encrypted traffic.

2 Proposed Method and Experiment

2.1 SPLT-SHAP Matching method



[Figure 1] Behavior-Level Visualization Result of SPLT-SHAP Matching (Top 2)

The proposed SPLT-SHAP matching method is a behavior-level interpretation technique that matches SHAP-derived feature explanations to the original SPLT sequence in encrypted traffic. Our method structures 75 SPLT features using three predefined attributes: Type (packet length | IAT (inter-arrival time)), (2) Direction (forward | backward | bidirectional), (3) Statistical property (max | min | mean | std). When SHAP identifies an important feature contributing to the classification result, the corresponding packets or intervals are traced according to these attribute conditions and visually matched to the original SPLT sequence.

2.2 Result of SPLT-SHAP matching

We collected encrypted traffic from YouTube and KakaoTalk identified via SNI (Server Name Indication), with 30,000 sessions per application. For the experiment, the dataset was split into training (70%) and testing (30%) sets, and we employed the XGBoost classifier, which achieved 99.3% accuracy on the test set using SPLT-based features.

An example of SPLT-SHAP matching is shown in [Figure 1]. According to the SHAP interpretation, the most influential feature was ACTIVE_MAX. This feature decomposes into three predefined attributes: (1) Type: IAT; (2) Direction: bidirectional; (3) Statistical Property: max. During feature extraction, the idle/active threshold was set to 100ms, which segments the packet sequence into alternating idle and active periods. The SPLT-SHAP matching algorithm then traces all active regions satisfying these attribute conditions and identifies the active interval with the maximum duration (0.235 secs). This enables analysts to directly pinpoint the exact behavioral evidence in the original SPLT plot that contributes to the AI model's prediction, and packet-level observables in encrypted traffic.

3 Conclusion and Future Work

In this paper, we proposed SPLT-SHAP matching for behavior-level interpretation of encrypted traffic classification. The method enables analysts to directly link feature explanations to observed traffic behavior. While this study focuses on YouTube and KakaoTalk sessions as proof-of-concept, the framework is designed to extend to various encrypted application classes. Future work includes: (1) validating the approach across diverse encrypted applications and datasets, (2) quantifying the fidelity of SHAP-to-SPLT mappings, and (3) integrating with SOC analyst workflows to measure operational benefits such as investigation time reduction.

4 References

[1] Anderson, B., & McGrew, D. (2016, October). Identifying encrypted malware traffic with contextual flow data. In Proceedings of the 2016 ACM workshop on artificial intelligence and security (pp. 35-46).

[2] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. In Advances in Neural Information Processing Systems (Vol. 30).